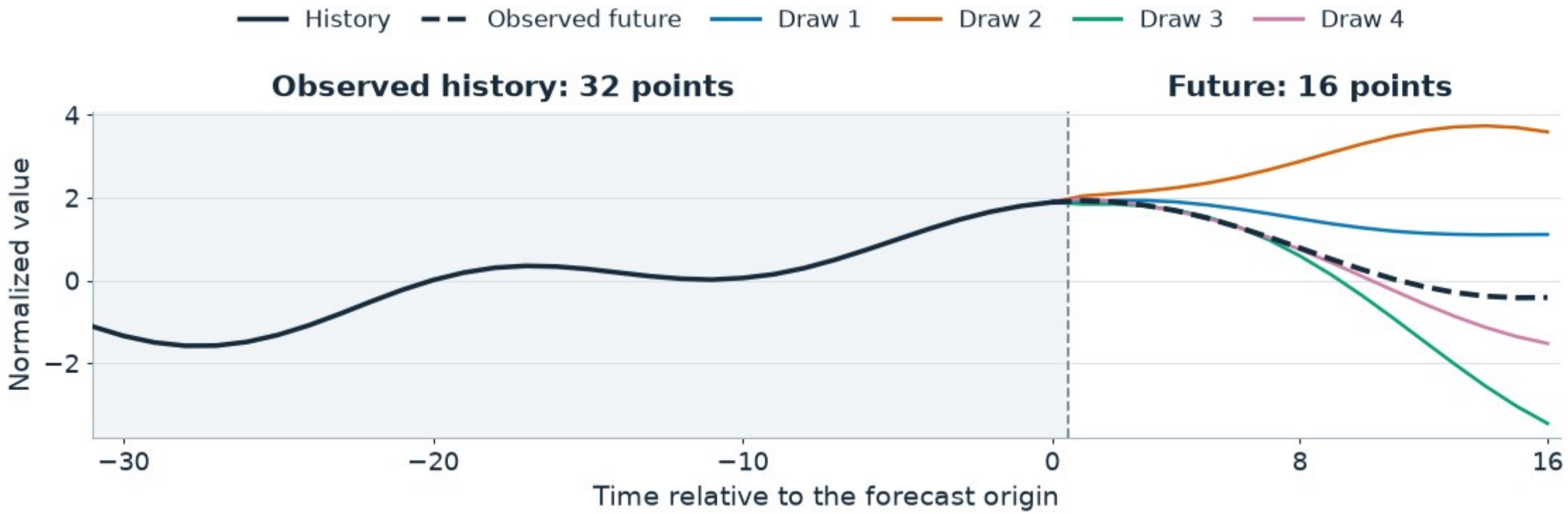
Smooth source, unstructured head

Source = starting noise; head = the forecasting network.
Does smooth noise need a network built for sequences?

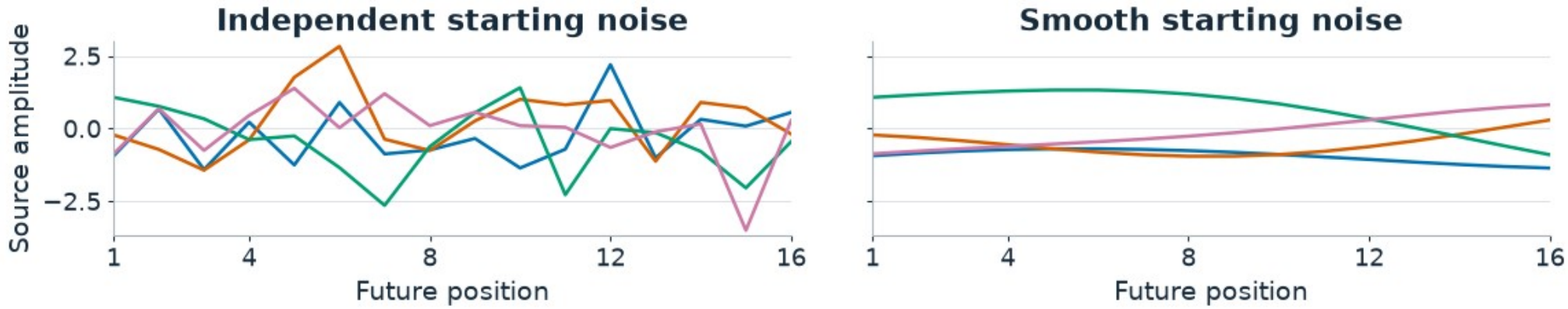
Daniil Koblov · Vasilii Lyamin · Egor Serov · Kirill Frolov
Sequential Data project · 26 October 2026

Forecasting a distribution of future trajectories



Synthetic data: a Gaussian process (GP), a distribution of smooth random curves.
Values = (value − history mean) / history standard deviation; time is in sample steps.
One noise draw gives one future trajectory; repeat draws to sample the distribution.

The network predicts velocity; integration gives a forecast



Training: noise ε, known future y → mixed curve x(t) = (1−t)ε + ty; target u = y−ε.

Network: current curve x, generation time t, past h → velocity v (rates of change).

Sampling: integrate dx/dt = v from noise to one future trajectory.

Generation time t runs from 0 to 1; it is not the time axis of the forecast.

Main experiment: two heads × two noise sources

Velocity network (head)	White noise	Smooth GP noise
MLP: dense network	5 training seeds	5 training seeds
S4: state-space model	5 training seeds	5 training seeds

MLP = multilayer perceptron; S4 = structured state-space sequence model.

6,000 updates per run; about 65,000 parameters. GP noise correlates future positions.

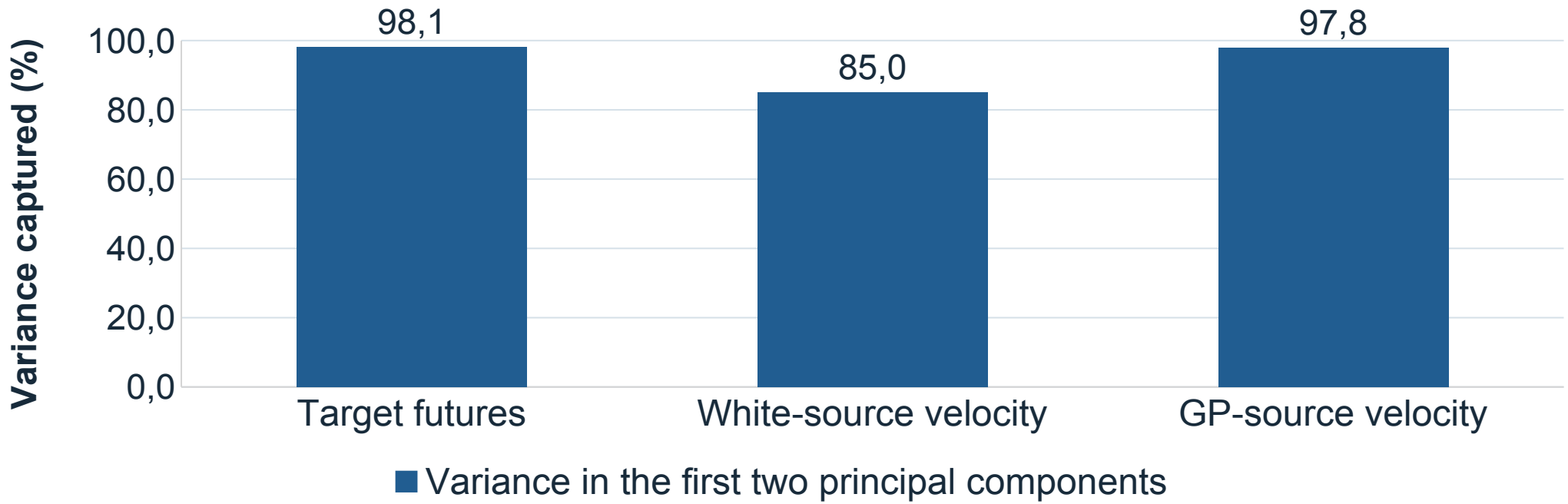
Whitened MLP (5 extra runs): rotate along directions of training-velocity variation;

divide by their standard deviations (with a floor), then undo the transform for scoring.

Hypothesis: with GP noise, S4 learns velocity directions with little variation better than MLP.

With GP noise, two directions contain 97.8% of velocity variance

PCA = principal component analysis: PC1, PC2, … rank directions by decreasing variance.
Separate train-fitted bases for 16-point futures y and velocities u = y−ε; bars use test data.



GP generates the data; white or GP noise starts the forecast. Values use history mean/std.
Design check: the first two velocity PCs should explain >90% of variance; GP passes.

GP-source residual error: MLP is lower than S4

MSE = mean squared error of predicted velocity v against target u (lower is better).
Errors use training-velocity PCs; residual = mean over PC3–16, after the two leading directions.

$$R_1(x) = \frac{1}{H-1} \sum_{i=1}^{H-1} \left| x_{i+1} - x_i \right|$$

x: one forecast trajectory
H: number of future points
i: forecast-point index

Reported ratio = mean R1 of generated paths / mean R1 of observed paths; 1 means a match.

Source / velocity head	MSE: PC1	Mean MSE: PC3–16	R1 ratio
WHITE / MLP	6.677	0.1221	1.058
WHITE / S4	7.321	0.1249	1.045
GP / MLP	14.929	0.0493	1.018
GP / S4	15.502	0.0583	1.039
GP / Whitened MLP	13.618	0.0401	1.016

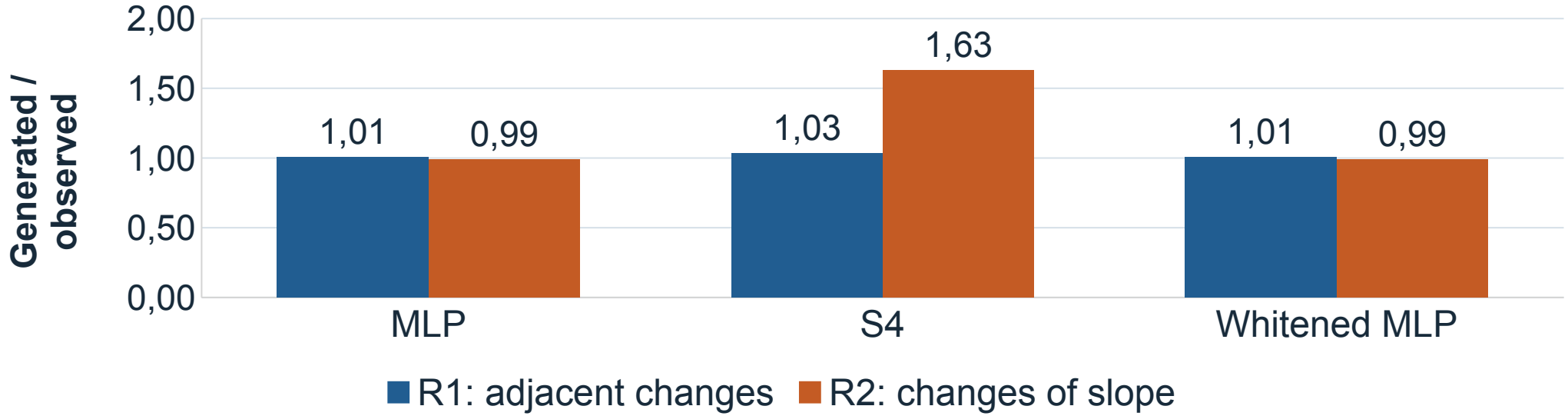
Original rule: mean MSE(PC3–16)/MSE(PC1) $\in$ [0.8,1.2] AND R1 ratio $\in$ [0.8,1.2].
Five-seed means. All 25 runs pass R1; 0/25 pass both conditions.

R1 misses changes in slope

$$R_2(x) = \frac{1}{H-2} \sum_{i=1}^{H-2} \left| x_{i+2} - 2x_{i+1} + x_i \right|$$

x: one forecast trajectory
H: number of future points
i: forecast-point index

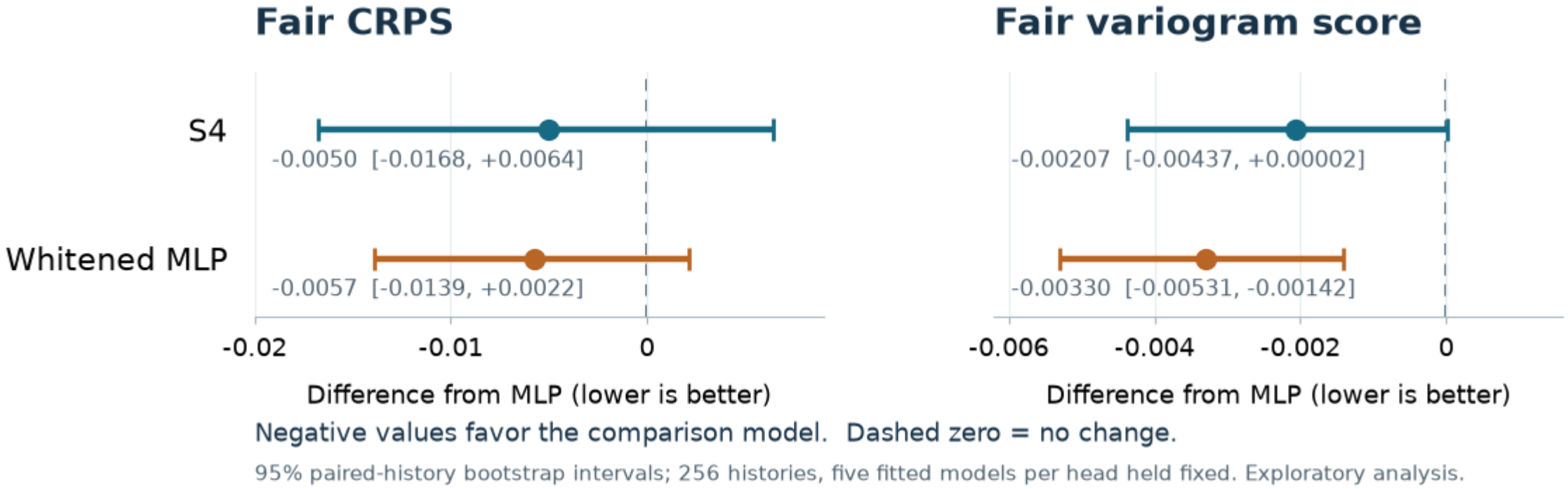
GP-source forecasts: 256 histories, 64 generated trajectories per history.



Each ratio = mean statistic of generated paths / mean statistic of observed paths.
S4 R2 ratio = 1.63. Greater curvature alone does not establish worse forecasts.

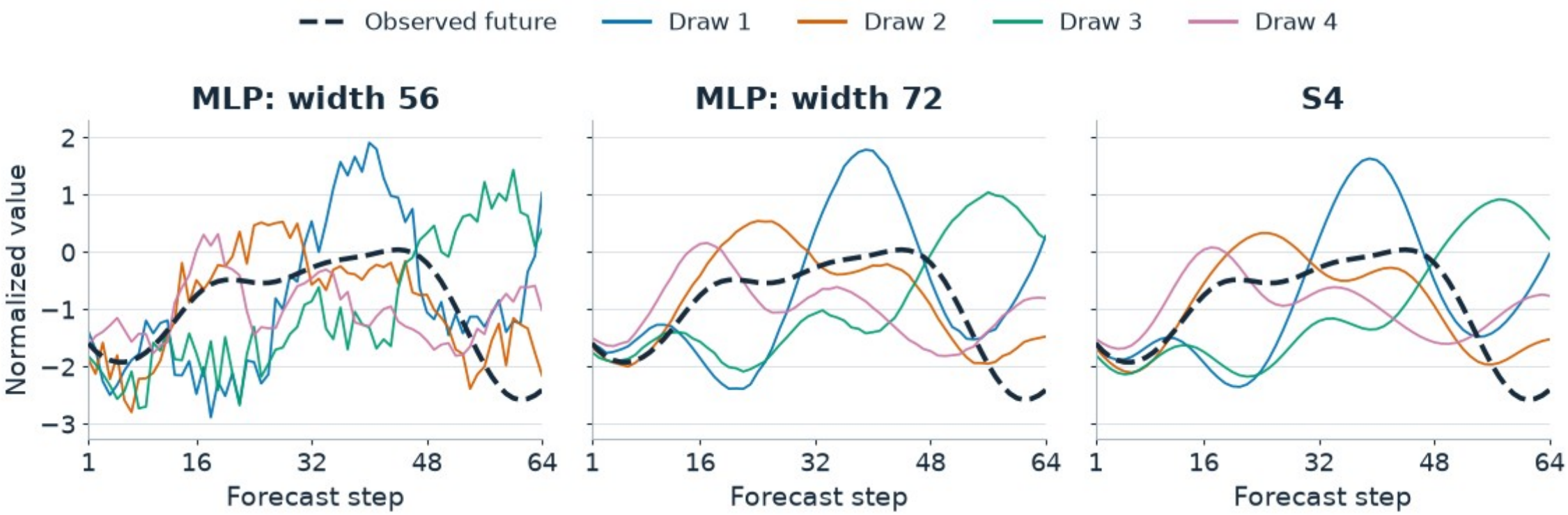
GP-source forecasts: shape and predictive scores differ

CRPS = continuous ranked probability score: predicted values at each future time.
VS = variogram score: predicted differences between times. Lower scores are better.
“Fair” corrects for the finite number of generated trajectories per history.



S4 intervals cross zero; whitening lowers VS. Negative differences favour that model.
95% confidence intervals resample histories while holding the fitted models fixed.

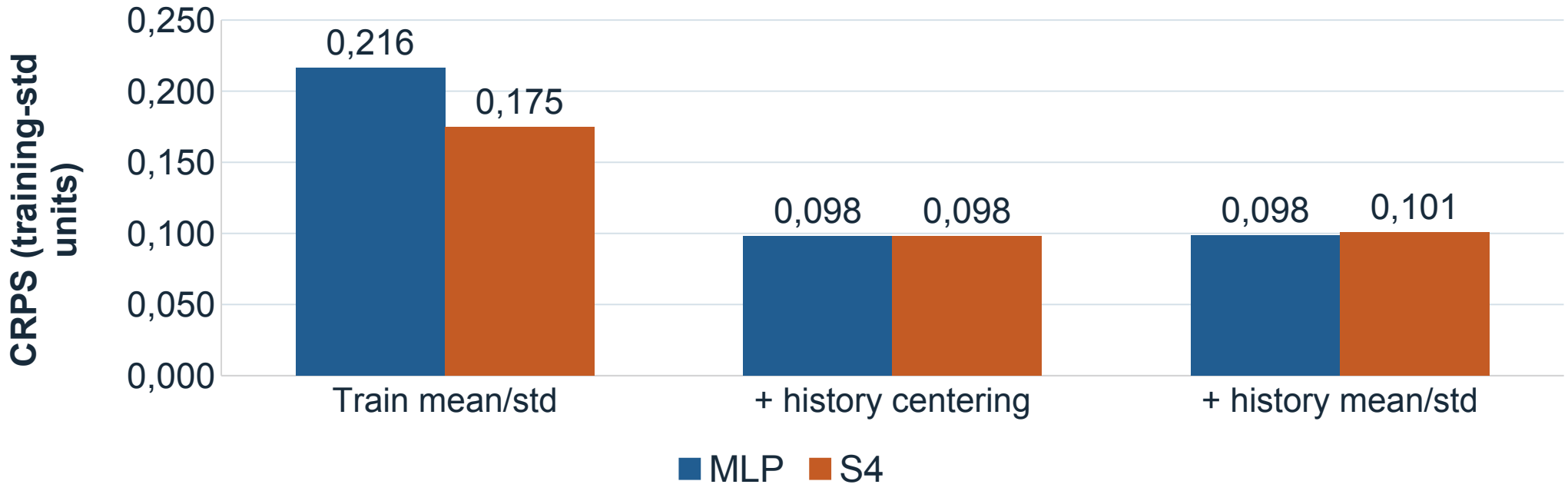
A wider MLP greatly reduces roughness at horizon 64



Separate check: 128 past / 64 future points, white noise, 12,000 updates.
Width = units in each MLP hidden layer; depth and the 64-value output stay fixed.
Across 3 new datasets, width 56 → 72 lowers mean R2 ratio from 19.09 to 1.60.

History centering sharply reduces the ETTh1 head gap

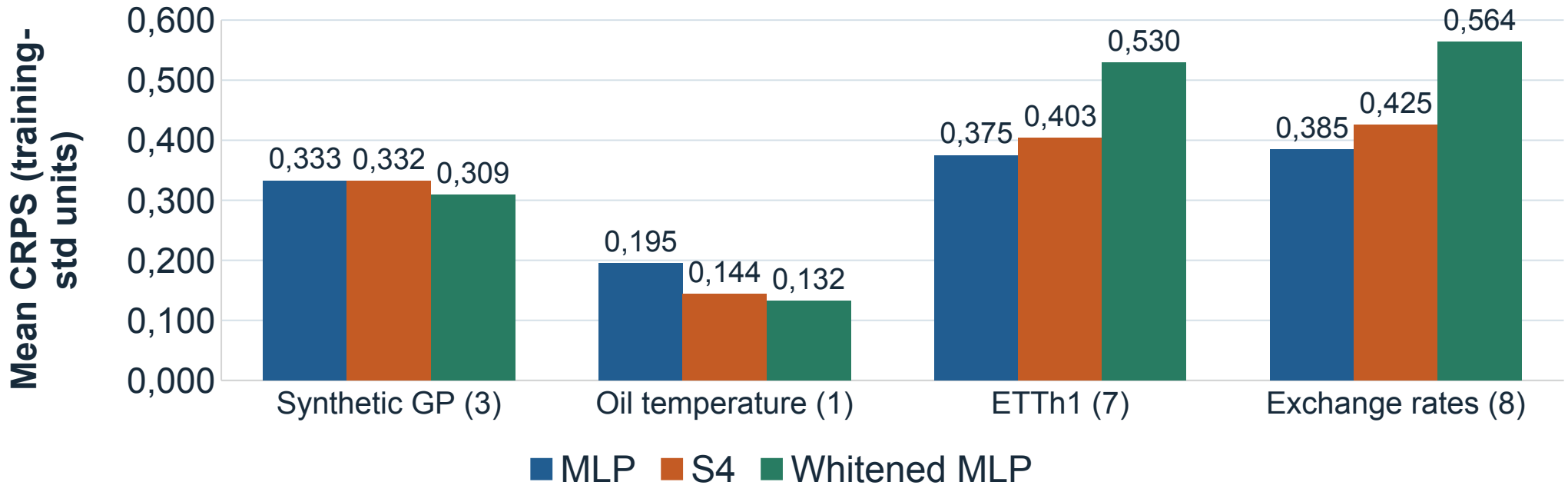
ETTh1: hourly transformer-oil temperature · 48 past / 24 future points · GP noise
6,000 updates; scores mapped back to training-standard-deviation units.



Centering subtracts the observed history mean; mean/std also divides by its standard deviation.
Ridge = regularized linear prediction + sampled training residuals, with history normalization.
Ridge CRPS: 0.0936 on the same 118 test windows.

Whitening helps synthetic GP, but hurts real multichannel forecasts

Channels = variables: GP simulations; hourly transformer measurements; daily exchange rates.
Each channel uses training mean/std; CRPS averages over forecast times and channels.



Energy score evaluates the whole future vector; lower is better and confirms the whitening trend.
Ridge + sampled training residual vectors: CRPS 0.193 / 0.093 / 0.287 / 0.068 (left to right).
Parentheses give channel counts. Compare heads within a task. GP noise; 2,000 training updates.

Checks prompted by the review

Review question	What the additional check shows
Are tiny PCA directions driving it?	GP velocity error on PC3–6: whitened MLP < MLP < S4
Does lower velocity error mean better forecasts?	GP forecasts: no resolved S4–MLP score gap; whitening lowers VS
Does the capacity effect repeat?	Wider MLP reduces roughness on 3 new synthetic datasets
Is the architecture comparison isolated?	Equal budgets; the heads still use history differently

Analytic reference = mean velocity given current x, time t and past h, from the known data law.
Broader per-head tuning and the whitening rescaling floor remain limitations.

Smooth starting noise alone does not justify switching to S4

Smooth GP noise did not produce the predicted S4 residual-error advantage.

Check both trajectory shape and forecast scores: they can tell different stories.

Capacity and normalization substantially change particular head comparisons.

Whitening helps some tasks, but hurts both real multichannel tasks tested here.

Use a simple linear baseline too. These are conclusions from fixed-budget experiments.

Team contributions

Team member	Contribution
Egor Serov	MLP and training pipeline; controlled comparisons; decoder width and capacity.
Kirill Frolov	Data and PCA; R2 / Variogram Score; forecast checks and final synthesis.
Daniil Koblov	S4 and structured heads; joint multichannel forecasting; transfer to real data.
Vasilii Lyamin	Whitening and normalization; Jacobian / kernel diagnostics; interpretation and plots.

Jacobian: velocity sensitivity to the current curve. Kernel: GP covariance function.
AI assisted implementation, analysis and writing; the authors remain responsible.

Backup: the literal success rule

Original rule: mean MSE(PC3–16) / MSE(PC1) must lie in [0.8, 1.2].

Generated R1 / target R1 must also lie in [0.8, 1.2].

All 25 runs pass the roughness part; 0 / 25 pass the full rule.

Earlier analytic reference also fails: it knows the generating law and the full raw history.

Keep the declared result. Diagnose why equality between unequal-variance directions is uninformative.

Backup: velocity errors restricted to PC3–6

Both columns average squared errors over PC3–6 of training velocities.
Left: $(v-u)^2$ against target $u=y-\varepsilon$. Right: $(v-v^*)^2$ against analytic mean $v^*=E[u|x,t,h]$.

GP-source head	Original velocity MSE	Squared gap to analytic field
MLP	0.172429	0.057731
S4	0.196926	0.076748
Whitened MLP	0.140330	0.022755

PC3–6 only. Original test: 8,192 windows; analytic comparison: 2,048 fresh windows.

Backup: averaged training labels improve forecasts modestly

Random labels: target $u=y-\varepsilon$ for one sampled future/noise pair.
Mean-velocity labels: average u over possible pairs at fixed curve x, time t and past h.

GP-source model	Fair CRPS ↓	R2 ratio	Fair VS ↓
MLP / random labels	0.4932	1.031	0.0714
MLP / mean-velocity labels	0.4785	1.028	0.0688
S4 / random labels	0.4923	1.649	0.0710
S4 / mean-velocity labels	0.4798	1.518	0.0688

3 paired dataset/seed runs. Mean-velocity labels use the known generating law.

Backup: variogram correction and uncertainty

X[m,i] = value at time i in generated trajectory m; M = number of trajectories. For times i,j: a[m] = |X[m,i]−X[m,j]|^0.5.

Time-pair weights are proportional to 1/|i−j| and sum to 1. Fair VS subtracts sample variance of a[m] divided by M for each pair.

95% confidence intervals resample 256 paired histories, holding 5 fitted models per head fixed.

Follow-ups: 3 paired dataset/training-seed runs; no pooled population confidence interval.

Post-hoc comparisons, no multiple-testing correction. Scores across different horizons are not directly comparable.